

Senior Design: Ethics Case Study

Homework No. 5
CHE 396- Senior Design
November 27, 2019

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Design Plan

Senior Design Ethics Case Presentation				
Task	Start Date	End Date	Duration (Days)	Assignee
Logistics	11/19	11/19	1	All
Tasks Assigned	11/19	11/19	1	All
Project Charter	11/20	11/20	1	All
Presentation	11/19	11/21	2	All
Power Point Slides	11/19	11/21	2	All
Report	11/22	11/27	6	All

Project Charter:

Title	Goal	In Scope	Out of Scope
Ethics Case Study	To present a study on an ethical case	• Studying an ethical case • Propose alternatives for approaching the case • Present benefits and consequences of the alternatives	
Deliverables		Team Members	
1. Stakeholders involved 2. Laws, guidelines or other standards govern 3. How problems or questions should be addressed 4. Who will be affected by this decision? 5. Alternatives to approach the issue 6. Benefits and consequences of those alternatives		• Huda Usman • Nisha Patel • Velia Palma • Ahmed Alghamdi	

Ethical Case: “Don’t Ask Don’t Tell” (equalization tank for the wastewater treatment plant)

“Reilly M. Karful, a wastewater process engineer with Slud, Gefl, Owsdown & Hill (SGOH), has spent the last few weeks designing an equalization tank for the wastewater treatment plant for the town of Whiteside. It so happens that the existing extended aeration treatment plant is located inside one room of a pre-engineered metal building, except for the headworks, which consist of a screen. The design that Reilly is proposing includes installing a new tank inside the treatment room and relocating the screen on top of the tank.

During a design review, the reviewer, Yvonne Moore-Karful (distant relation to Reilly) of the architectural firm Plansem & Drawsem, notices that sewage plant headworks are covered by National Fire Protection Association (NFPA) Code Section 820, which requires certain areas to be Class I, Division 1 or Class I, Division 2; i.e., “explosion proof.” Concerned about the potential problem, Yvonne calls to let Reilly know that moving the screen into the treatment room would require the room to be classified as explosion proof, and she follows up this call with a memo and her notations on the design drawings. Reilly decides to double-check this information and he reviews NFPA 820 himself. The code indeed appears to require the headworks and the existing treatment room to be explosion proof because the plant does not have primary settling tanks.

Reilly informs the project manager at SGOH, Dante McWaves, about the matter, and gives him a copy of Yvonne’s memo. With Reilly in his office, Dante phones the client, Bull Parker, the town commissioner, to make him aware of Reilly’s concerns. Bull happens to be cleaning his shotgun when he gets Dante’s call. News of cost increases and delays does not please Bull, and he pointedly reminds Dante that the project is behind-schedule, that Whiteside needs this addition to the wastewater treatment plant to avoid non-compliance with State regulations, and that Whiteside is not a wealthy town. As he trips the action on the shotgun, as if to place a shell in the chamber (Dante hears this over the phone), Bull tells Dante that he’ll have the County Fire Code Official, Bobby Burns, get back to Dante.

Reilly notices the color drain from Dante’s face and that his hands visibly shake as he hangs up the phone. Within minutes, the fire code official calls and informs Dante that the existing room is already explosion proof. What luck! The problem is

solved and Dante, still pale yet with obvious relief, informs Reilly that he doesn't want to bother Bull with any more problems. Dante then clearly and unequivocally directs Reilly to finish the project.

But Reilly is concerned. While Bobby Burns' statement satisfies Dante, Reilly suspects the fire code official could be wrong. Though Reilly has limited experience with fire codes, he has noticed that the electrical sockets, light switches, lighting fixtures, and junction boxes appear normal and don't give the appearance of being explosion proof. While part of him wants to pursue the matter, he also realizes that exploring his doubts about the existing room being explosion proof will not only further stress out Dante, but could result in additional project delays, in Whiteside having to fund an expensive retrofit to bring their existing facility up to code, and in further annoying Bull.

What, if anything, should Reilly do?" [1]

The engineers are hired by variety of companies based on their capabilities and to perform to their best of the knowledge. It is the most important responsibility of an engineer to provide safe and reliable products. It is part of the ethic code that they avoid conflicts, keep confidentiality in company matters and be professional. However, there are variety of cases where the code of ethics for an engineer is compromised. Therefore, while performing jobs, engineers should remind themselves every day that the decision they take affects health, safety, and welfare of their community.

In this case study, the main ethical issue is making certain that the fire codes are fully adhered to and ensuring the safety of everyone at the waste treatment plant even if it means going against the perceived wishes of Reilly's manager and client, the town commissioner.

There are several other issues that can be considered ethical concern in this case. First, the reviewer, Yvonne Moore-Karful is a distant relation to Reilly. In this instance no one seems to raise concern about their familial relationship, but it could be brought up as an issue if something went wrong in the future. Reilly could be blamed for hiring Yvonne not based on merit but for personal reasons. Second, after informing the town commissioner about the additional time and money that might need to be invested in order to ensure the room is explosion proof and adhering to the fire code, Reilly and her manager are left with the impression that they are being pressured or threatened to complete their work and ignore this fire code. Their perception of this is confirmed when Dante, Reilly's manager, receives a phone call from the fire official of the town and is informed that the room is fire proofed and the project should continue. While Reilly has no proof that the fire official is being unethical and neglecting her duties, the prompt response after raising her concerns makes her think the fire official might be acting on the direction of the commissioner in order to not delay the

project. Finally, Reilly is left to decide whether to continue with the project or look further into explosion proofing the room.

It is important to note that Reilly is not familiar with what it takes to explosion proof a room but because all the fixtures and outlets look conventional, she has reservations that the fire official is approving the project without proper inspection. This leaves Reilly to consider how to move forward and address the situation in the most effective and ethical manner possible.

To make a decision for how to proceed with a project when a challenge is encountered, one must consider factors such as the project finances, laws and regulations, the project timeline, and the wishes of the client as well as in this case the opinion of supervisors and/or managers. In this case any decision made will have an effect on all of the factors above. If Reilly decides to investigate further and the room need to be explosion proofed this will mean more money needs to be invested and that more time will be needed to finish the project. If the room is not fire proofed but the project continues as is then the project will be in violation of the National Fire Protection Association (NFPA) Code, Section 820 and this could mean in the event of an explosion the workers of the waste water treatment plant could be put in life threatening danger. In addition to all this, the perceived threat and pushback from the town commissioner means that Reilly and her company would have to deal with an unhappy unsatisfied customer because he is responsible for making sure the people of Whiteside have clean water and need to wastewater treatment plant to be up and running as soon as possible.

This case will not only affect the company itself but also the town of white side. If the room will not be built explosion proof, the reputation of Bull Parker, as a town commissioner, will be at stake as well.

There always comes a point in an engineers' life that they have to make these tough decisions. We have to evaluate which route to take and what will be the most ethical way to approach the issue. In order to do so we have to use the most appropriate source such as AIChE code of ethics. For this particular case, the following code of ethic is appropriate: "Hold paramount the safety, health and welfare of the public and protect the environment in performance of their professional duties." If the treatment room was not made sure to be explosion proof and something went wrong during the wastewater treatment and explosion occurred, residents of the Whiteside town will be in danger. Also, the explosion will cause emission of hazardous gases which will adversely affect the environment. The code of ethic also stated that the member should "Conduct themselves in a fair, honorable and respectful manner" which will allow them to have a good reputation in the company they were hired to do the job for.

While there may seems to be many factor pushing Reilly to overlook her concerns and move along with the project it is important that she considers the code of ethics which has been established in the interest of the public and the environment. Because her manager, the commissioner and the fire officer have given her instructions to move ahead and build the tank in the room as it currently Reilly is left having to disproof her concern on her so that she can be sure the room is in compliance with all the codes and safe for the moving the tank in. One way she can go about addressing her concerns is by bringing in a third party to inspect tube room. In

the event that the room needs to be explosion proofed, Reilly will then need to make the appropriate changes to her design and secure the capital from the town of Whiteside. This decision will ensure that all of the codes and regulations have been adhered to the best of Reilly's knowledge. She can use the inspection to gain experience about fireproof code. This event will provide an opportunity for her to show the company that she can be trusted to deliver quality work and that adhere to all standards.

However this decision could have unfavorable results for Reilly. In the event that the room was already fire proofed, as the fire official and commissioner had said then Reilly would have made an incorrect and stubborn decision in the eyes of her manager and employees. She could be seen as difficult to work with or not a team player. This would also mean that Reilly brought a third party evaluator in for no reason and increased the cost of the project when there was no need.

Another alternative course of action that Reilly should take is by approaching directly to the source i.e. the National Fire Protection Association (NFPA) Department. She should give the situation and her manager Dante some time like a day or two, and approach him again. She should express her concerns for the situation and the convenient statement made by the fire code official on the room being explosion proof few minutes right after Dante got done talking with town commissioner Bull Parker. Furthermore, Reilly should also suggest Dante that both should go together to visit the fire code official in person to fix an appointment with him to visit the plant and the treatment room in person for an inspection. This can not only clear the doubt of fire code official lying but also it clears the doubt that if he is not lying, is he mentioning about the right treatment room and not some other room in the plant which has already been explosion proofed.

The upside of this alternative is that the first AIChE Code of Ethics i.e. "Hold paramount the safety, health and welfare of the public and protect the environment in performance of their professional duties." is followed [2]. As an engineer, its one's paramount responsibility to look after the safety of their surroundings which includes, co-workers, civilians, and especially the environment and thereby following the 2nd code of ethic "Formally advise their employers or clients (and consider further disclosure, if warranted) if they perceive that a consequence of their duties will adversely affect the present or future health or safety of their colleagues or the public." [2] When Reilly let some time pass by for her manager Dante to calm down and then approach about the situation again in a calm manner, it shows her quality of a sensible and responsible engineer. She will be making sure the safety of her coworkers along with saving company from facing any future accidents which could lead to a great loss to the company. By approaching the manager again with an alternative solution to ensure the safety of that treatment room, she is making sure that she is being respectful of the work assigned to her by her manager and also following the company's client interest and not conflicted with their decisions and thus following the 5th code of ethic is met i.e. "Act in professional matters for each employer or client as faithful agents or trustees, avoiding conflicts of interest and never breaching confidentiality." Altogether, by following this course of steps to get the treatment plant explosion proof, Reilly will build her reputation at work and also will help rise her self-esteem as she will be performing her duties in fair and honest manner and thus following the 8th and 11th code of ethic.

The downside of this alternative will be that the total project cost will face a significant rise as performing and declaring the entire treatment room explosion proof requires labor, special equipment and replacements, etc. The rise in the project cost will ultimately impact the budget of the Whiteside Town which is not in a good financial state. Furthermore, conducting the explosion proof agenda will delay the project completion which won't impress the town commissioner Bull Parker and can create a great tension between the commissioner and the company and directly to Dante as he talked with the commissioner directly on the call and it seemed that the problem was solved right away.

References:

1. National Institute for Engineering Ethics (NIEE), Texas Tech University. "Case 1030: Don't Ask, Don't Tell..." *Murdough Center for Engineering Professionalism*.
2. Code of Ethics." AIChE The Global Home of Chemical Engineers, November 2015.